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ON A MIXED FINITE ELEMENT METHOD FOR THE p-LAPLACIAN

MOHAMED FARHLOUL AND HASSAN MANOUZI

ABSTRACT. In this article we propose a mixed method for the p-Laplacian problem. We improve the result obtained in [**6**] by showing that the error estimates obtained, for a mixed finite element method, are valid in three dimensions without restrictions. The key of this improvement is the use of a more appropriate functional setting. This allows also to improve the previously known error estimates.

1. Introduction. Let Ω be a bounded open set of $\mathbf{R}^N$, $N = 2$ or 3, with a Lipschitz boundary Γ. The p-Laplace equation is the quasilinear (degenerate) partial differential equation:

$$\begin{cases} -\nabla \cdot (|\nabla u|^{p-2}\nabla u) = f & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma, \end{cases} \tag{1.1}$$

where $1 < p < \infty$, ∇u is the vector gradient of u, and

$$|\nabla u| = \left(\sum_{i=1}^{N} \left(\frac{\partial u}{\partial x_i} \right)^2 \right)^{1/2}.$$

This type of equation appears in many mathematical models of physical processes: nonlinear diffusion and filtration, see Philip [**14**]; power-law materials, see Atkinson and Champion [**1**]; and viscoelastic materials, see Le Tallec [**9**], for example.

The approximation of (1.1) by finite elements has been studied in several works (see Glowinski and Marrocco [**8**], Ciarlet [**5**], Tyukhtin [**19**] and Barrett and Liu [**3**]). Further results for more general problems, including (1.1), are given in Liu and Barrett [**10**], [**11**].

In Farhloul [**6**] we have studied a mixed finite element method for problem (1.1) in the case $1 < p < 2$. This mixed finite element method is based on the introduction of $\sigma = |\nabla u|^{p-2}\nabla u$ as a new unknown and its approximation by the lowest degree Raviart-Thomas finite element

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(cf. Thomas [**18**], Raviart and Thomas [**15**] and Nedelec [**13**]). We have proved in [**6**] error estimates in both two and three dimensions but in three dimensions we had a restriction on p, $6/5 \leq p < 2$ (see Theorem 3.2 in [**6**]).

It is the purpose of this paper to give a proof of error estimates which are valid in three dimensions without restrictions on p, i.e., for all $p > 1$. The basic idea is to replace the functional setting in [**6**] by a more appropriate one. More precisely, in the case $1 < p < 2$, we prove an optimal error bound for u, and we improve the one obtained in Baranger et al. [**2**] for σ. Furthermore, we extend our analysis to the case $p > 2$ by proving the same error bound for u as the one obtained in Glowinski and Marrocco [**8**] and improving the one obtained in Glowinski and Marrocco [**8**] for σ (see also Ciarlet [**5**]). We point out that all these estimates are obtained under minimal regularity of the solution of problem (1.1) (compare with the results obtained in Barrett and Liu [**3**], and Theorem 3.2 in [**6**]). Finally, for some numerical results for a nonlinear Stokes model using the mixed finite element proposed here, we refer to Manouzi and Farhloul [**12**].

2. Mixed formulation. The mixed formulation is based on the inversion of the relation $\sigma = |\nabla u|^{p-2}\nabla u$, i.e.,

$$\nabla u = |\sigma|^{q-2}\sigma$$

where $(1/p) + (1/q) = 1$. With this purpose the problem (1.1) is decomposed into a first order system as follows

$$\left\{\begin{array}{rl} |\sigma|^{q-2}\sigma = \nabla u & \text{in } \Omega, \\ \nabla \cdot \sigma + f = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma. \end{array}\right. \tag{2.1}$$

The weak formulation appropriate for the mixed method is

$$\left\{\begin{array}{l} \displaystyle\int_\Omega A(\sigma)\cdot\tau\,dx + \int_\Omega \nabla\cdot\tau u\,dx = 0 \quad \forall\,\tau \in X, \\ \displaystyle\int_\Omega \nabla\cdot\sigma v\,dx + \int_\Omega f v\,dx = 0 \quad \forall\, v \in M, \end{array}\right. \tag{2.2}$$

where

$$A(\sigma) = |\sigma|^{q-2}\sigma, \tag{2.3}$$

$$X = \{\tau \in (L^q(\Omega))^N;\ \nabla \cdot \tau \in L^q(\Omega)\}, \quad M = L^p(\Omega), \tag{2.4}$$

and $f \in L^q(\Omega)$.

We denote the norm of X by $\|\tau\|_X = (\|\tau\|^q_{0,q,\Omega} + \|\nabla\cdot\tau\|^q_{0,q,\Omega})^{1/q}$, where $\|\tau\|_{0,q,\Omega} = (\int_\Omega |\tau|^q\, dx)^{1/q}$, and the norm of M by $\|v\|_M = \|v\|_{0,p,\Omega}$.

Remark 2.1. The difference between the functional setting here and the one in [**6**] is that in [**6**] we have considered $\nabla \cdot \tau \in L^2(\Omega)$ and $M = L^2(\Omega)$.

To prove the existence and the uniqueness of the solution of (2.2), we shall use the following abstract result (see [**7**, p. 61, Remark 4.2], Sandri [**17**]).

Lemma 2.1. *Let $(X, \|\cdot\|_X)$ and $(M, \|\cdot\|_M)$ be two reflexive Banach spaces. Let $(X', \|\cdot\|_{X'})$ and $(M', \|\cdot\|_{M'})$ be their corresponding dual space. Let $B : X \to M'$ be a linear continuous operator and $B' : M \to X'$ the dual operator of B. Let $V = ker(B)$ be the kernel of B; we denote by $V^0 \subset X'$ the polar set of V, i.e., $V^0 = \{x' \in X', \langle x', v\rangle = 0, \forall\, v \in V\}$ and $\dot{B} : (X/V) \to M'$ the quotient operator associated with B. The three following properties are equivalent*:

(i) *$\beta > 0$ exists such that*

$$\inf_{r\in M} \sup_{v\in X} \frac{\langle Bv, r\rangle}{\|r\|_M \|v\|_X} \geq \beta,$$

(ii) *B' is an isomorphism from M onto V^0 and*

$$\|B'r\|_{X'} \geq \beta\|r\|_M \quad \forall\, r \in M,$$

(iii) *$\dot{B}$ is an isomorphism from (X/V) onto M' and*

$$\|\dot{B}\dot{v}\|_{M'} \geq \beta\|\dot{v}\|_{(X/V)} \quad \forall\, \dot{v} \in (X/V).$$

Now let us prove the following result concerning the inf-sup condition.

Proposition 2.1. *There exists a positive constant β such that*

$$\inf_{v\in M}\sup_{\tau\in X}\frac{\int_\Omega \nabla\cdot\tau v dx}{\|v\|_M\|\tau\|_X}\geq\beta. \tag{2.5}$$

Proof. Let $v\in M$ and w denote the solution of the p-Laplace problem

$$\begin{cases} -\nabla\cdot(|\nabla w|^{p-2}\nabla w)=|v|^{p-2}v & \text{in } \Omega,\\ w=0 & \text{on } \Gamma. \end{cases}$$

From Ciarlet [**5**] and Glowinski and Marrocco [**8**], we know that this problem has a unique solution $w\in W_0^{1,p}(\Omega)$ and there exists a constant C such that

$$\|w\|_{1,p,\Omega}\leq C\|v\|_{0,p,\Omega},$$

where

$$\|w\|_{1,p,\Omega}=(\|w\|_{0,p,\Omega}^p+\|\nabla w\|_{0,p,\Omega}^p)^{1/p}.$$

Therefore, $\tau^*=-|\nabla w|^{p-2}\nabla w$ is in the space X and there exists a constant C such that

$$\begin{aligned}\|\tau^*\|_X^q&=\|\tau^*\|_{0,q,\Omega}^q+\|\nabla\cdot\tau^*\|_{0,q,\Omega}^q\\&=\|\nabla w\|_{0,p,\Omega}^p+\|v\|_{0,p,\Omega}^p\\&\leq C\|v\|_{0,p,\Omega}^p.\end{aligned}$$

Hence $\|\tau^*\|_X\leq C\|v\|_{0,p,\Omega}^{p-1}$, and $\int_\Omega\nabla\cdot\tau^* v\,dx=\|v\|_{0,p,\Omega}^p$. Thus, (2.5) follows immediately. □

As a consequence of Lemma 2.1 and Proposition 2.1, we have the following result.

Theorem 2.1. *Problem* (2.2) *has a unique solution* $(\sigma,u)\in X\times M$ *and there exists a constant* $C>0$ *such that*

$$\|\sigma\|_X+\|u\|_M\leq C(\|f\|_{0,q,\Omega}+\|f\|_{0,q,\Omega}^{q-1}).$$

Proof. The proof of this result is the same as the one of Theorem 2.1 in [**6**]. □

3. Discrete problem and error analysis. Let T_h be a triangulation of $\overline{\Omega}$ into closed triangles (in the case $N = 2$) or tetrahedrons (in the case $N = 3$). The elements of T_h are assumed to be regular in the usual sense (cf. Ciarlet [**5**]). We define the finite element spaces

$$X_h = \{\tau_h \in X; \tau_h|_K \in RT_0(K), \forall\, K \in T_h\}, \tag{3.1}$$

$$M_h = \{v_h \in M; v_h|_K \in P_0(K), \forall\, K \in T_h\}, \tag{3.2}$$

where $P_0(K)$ denotes the space of constants and

$$RT_0(K) = (P_0(K))^N + xP_0(K), \quad x = (x_1, \dots, x_N).$$

The finite element space X_h is nothing else than the approximation of the space X by the lowest degree element of Raviart-Thomas (cf. Raviart and Thomas [**15**], Nedelec [**13**], Brezzi and Fortin [**4**], Roberts and Thomas [**16**]).

Then the finite element approximation to problem (2.2) is given by $(\sigma_h, u_h) \in X_h \times M_h$ satisfying

$$\begin{cases} \displaystyle\int_\Omega A(\sigma_h) \cdot \tau_h \, dx + \int_\Omega \nabla \cdot \tau_h u_h \, dx = 0 \quad \forall\, \tau_h \in X_h, \\ \displaystyle\int_\Omega \nabla \cdot \sigma_h v_h \, dx + \int_\Omega f v_h \, dx = 0 \quad \forall\, v_h \in M_h. \end{cases} \tag{3.3}$$

We introduce the Raviart-Thomas projection (cf. [**13**], [**15**]), Π_h defined from $X \cap (W^{1,s}(\Omega))^N$, $s > 1$, onto X_h by

$$\int_e (\tau - \Pi_K \tau) \cdot n \, ds = 0 \quad \forall\, e \in \partial K, \forall\, K \in T_h \tag{3.4}$$

where $\Pi_K \tau = \Pi_h \tau|_K$ and n denotes the unit outward normal to the boundary ∂K of K.

The operator Π_h is well defined for all $\tau \in (W^{1,s}(\Omega))^N$, $s > 1$. Indeed, for every $K \in T_h$, $\tau|_K \in (W^{1,s}(K))^N$ and thus $\tau|_{\partial K} \in$

$(W^{1-1/s,s}(\partial K))^N \subset (L^s(\partial K))^N \subset (L^1(\partial K))^N$. Therefore, $\int_e \tau \cdot n \, ds$ has sense for every $e \subset \partial K$.

Moreover (cf. [**4**], [**6**], [**13**], [**15**]),

(3.5)
$$\int_\Omega \nabla \cdot (\Pi_h \tau - \tau) v_h \, dx = 0, \quad \forall \tau \in X \cap (W^{1,s}(\Omega))^N, \ \forall v_h \in M_h,$$

(3.6) $$\|\tau - \Pi_h \tau\|_{0,s,\Omega} \leq Ch|\tau|_{1,s,\Omega}, \quad \forall \tau \in (W^{1,s}(\Omega))^N, \ s > 1.$$

Remark 3.1. In the case $q > 2$, i.e., $1 < p < 2$, the operator Π_h is well defined for all $\tau \in X$ (cf. Brezzi and Fortin [**4**]) and (3.5) is valid for all $\tau \in X$.

Before giving the error estimates for problem (3.3), let us first prove the following result.

Proposition 3.1. *There exists a constant C independent of h such that*

(3.7) $$\inf_{v_h \in M_h} \sup_{\tau_h \in X_h} \frac{\int_\Omega \nabla \cdot \tau_h v_h dx}{\|v_h\|_M \|\tau_h\|_X} \geq C.$$

Proof. Let $v_h \in M_h$ and $\tau^* = -|\nabla w|^{p-2} \nabla w$ where w is the solution of the p-Laplace problem

$$\begin{cases} -\nabla \cdot (|\nabla w|^{p-2} \nabla w) = |v_h|^{p-2} v_h & \text{in } \Omega, \\ w = 0 & \text{on } \Gamma. \end{cases}$$

This problem has a unique solution $w \in W_0^{1,p}(\Omega)$, and there exists a constant $C > 0$ such that $\|w\|_{1,p,\Omega} \leq C \|v_h\|_{0,p,\Omega}$. Thus, $\tau^* \in X$ and $\|\tau^*\|_X \leq C \|v_h\|_{0,p,\Omega}^{p-1}$. Now, let $\tilde{\tau} = \nabla \tilde{w}$, where $\tilde{w}$ is the solution of the Dirichlet problem

$$\begin{cases} \Delta \tilde{w} = \nabla \cdot \tau^* & \text{in } \Omega, \\ \tilde{w} = 0 & \text{on } \Gamma. \end{cases}$$

Since $\nabla \cdot \tau^* \in L^q(\Omega)$, we have $\tilde{\tau} \in (W^{1,q}(\Omega))^N$, there exists a constant C such that $\|\tilde{\tau}\|_{1,q,\Omega} \leq C \|\nabla \cdot \tau^*\|_{0,q,\Omega}$, and $\nabla \cdot \tilde{\tau} = \nabla \cdot \tau^*$ in Ω.

Finally, let $\tilde{\tau}_h = \Pi_h \tilde{\tau}$. Due to (3.5) and (3.6), we have

$$\int_\Omega \nabla \cdot \tilde{\tau}_h v_h \, dx = \int_\Omega \nabla \cdot \tilde{\tau} v_h \, dx = \int_\Omega \nabla \cdot \tau^* v_h \, dx = \|v_h\|_{0,p,\Omega}^p$$

and

$$\|\tilde{\tau}_h\|_X \le C(\|\tilde{\tau}\|_{1,q,\Omega} + \|\nabla \cdot \tilde{\tau}\|_{0,q,\Omega}) \le C\|\nabla \cdot \tau^*\|_{0,q,\Omega} \le C\|v_h\|_{0,p,\Omega}^{p-1}.$$

Therefore, (3.7) follows immediately. □

Due to (3.7), the existence and uniqueness of the solution of problem (3.3) are obtained as in Theorem 2.1.

We are now in a position to state the error estimates for problem (3.3). We begin by the case $2 \le p < \infty$.

Theorem 3.1. *Suppose that $2 \le p < \infty$. Let (σ, u) be the solution of (2.2) and (σ_h, u_h) the solution of (3.3). If $(\sigma, u) \in (W^{1,q}(\Omega))^N \times W^{1,p}(\Omega)$, then there exists a constant C, independent of h, such that*

$$\|\sigma - \sigma_h\|_{0,q,\Omega} \le Ch^{q/2}, \tag{3.8}$$

$$\|u - u_h\|_{0,p,\Omega} \le Ch^{q-1}, \tag{3.9}$$

where $(1/p) + (1/q) = 1$.

Proof. To prove the error estimate (3.8), first observe that, from the second equation of (2.2) and of (3.3), respectively, and (3.5),

$$\int_\Omega \nabla \cdot (\Pi_h \sigma - \sigma_h) v_h \, dx = 0 \quad \forall\, v_h \in M_h.$$

On the other hand, by using the first equation of (2.2) and (3.3), respectively, and this last relation, we get

$$\int_\Omega (A(\sigma) - A(\sigma_h)) \cdot (\sigma_h - \Pi_h \sigma) \, dx = 0,$$

and then

$$\int_\Omega (A(\sigma) - A(\sigma_h)) \cdot (\sigma - \sigma_h) \, dx = \int_\Omega (A(\sigma) - A(\sigma_h)) \cdot (\sigma - \Pi_h \sigma) \, dx. \tag{3.10}$$

Now, using the fact that (cf. Sandri [**17**])

$$\int_\Omega (A(\sigma)-A(\sigma_h))\cdot(\sigma-\sigma_h)\,dx \geq C\frac{\|\sigma-\sigma_h\|^2_{0,q,\Omega}}{\|\sigma\|^{2-q}_{0,q,\Omega}+\|\sigma_h\|^{2-q}_{0,q,\Omega}} + C\int_\Omega |A(\sigma)-A(\sigma_h)||\sigma-\sigma_h|\,dx,$$

$$\int_\Omega (A(\sigma)-A(\sigma_h))\cdot(\sigma-\Pi_h\sigma)\,dx \leq C\left[\int_\Omega |A(\sigma)-A(\sigma_h)||\sigma-\sigma_h|\,dx\right]^{1/p}\|\sigma-\Pi_h\sigma\|_{0,q,\Omega},$$

the inequality of Young: for $\varepsilon>0$, $a\geq 0$ and $b\geq 0$,

$$ab \leq \frac{1}{p}(\varepsilon a)^p + \frac{1}{q}\left(\frac{b}{\varepsilon}\right)^q,$$

and (3.10), we get

$$\frac{\|\sigma-\sigma_h\|^2_{0,q,\Omega}}{\|\sigma\|^{2-q}_{0,q,\Omega}+\|\sigma_h\|^{2-q}_{0,q,\Omega}} + \int_\Omega |A(\sigma)-A(\sigma_h)||\sigma-\sigma_h|\,dx \leq C\|\sigma-\Pi_h\sigma\|^q_{0,q,\Omega}. \tag{3.11}$$

Thus,

$$\|\sigma-\sigma_h\|^2_{0,q,\Omega} \leq C(\|\sigma\|^{2-q}_{0,q,\Omega}+\|\sigma_h\|^{2-q}_{0,q,\Omega})\|\sigma-\Pi_h\sigma\|^q_{0,q,\Omega}.$$

This last estimate, with (3.6), gives us

$$\|\sigma-\sigma_h\|_{0,q,\Omega} \leq Ch^{q/2}|\sigma|^{q/2}_{1,q,\Omega},$$

which is the desired result.

It remains to prove the error estimate (3.9). First, observe that, from the first equation of (2.2) and (3.3), respectively,

$$\int_\Omega \nabla\cdot\tau_h(P_hu-u_h)\,dx = \int_\Omega (A(\sigma_h)-A(\sigma))\cdot\tau_h\,dx \quad \forall\,\tau_h\in X_h,$$

where $P_h u$ is the projection of u onto $\Pi_{K\in T_h} P_0(K)$.

Thus, using this last relation and the inf-sup condition (3.7), we get

$$\begin{aligned} C\|P_h u - u_h\|_{0,p,\Omega} &\le \sup_{\tau_h\in X_h} \frac{\int_\Omega \nabla\cdot\tau_h(P_h u - u_h)dx}{\|\tau_h\|_X} \\ &\le \sup_{\tau_h\in X_h} \frac{\int_\Omega (A(\sigma_h) - A(\sigma))\cdot\tau_h dx}{\|\tau_h\|_X}. \end{aligned}$$

Therefore,

$$\|P_h u - u_h\|_{0,p,\Omega} \le C\bigg[\int_\Omega |A(\sigma) - A(\sigma_h)||\sigma - \sigma_h|\,dx\bigg]^{1/p}$$

due to the fact that (cf. Sandri [**17**]) for all $\tau_h \in X_h$,

$$\begin{aligned} &\int_\Omega (A(\sigma) - A(\sigma_h))\cdot\tau_h\,dx \\ &\qquad\le C\bigg[\int_\Omega |A(\sigma) - A(\sigma_h)||\sigma - \sigma_h|\,dx\bigg]^{1/p}\|\tau_h\|_{0,q,\Omega}. \end{aligned}$$

On the other hand, by using (3.11) and this last estimate, we get

$$\|P_h u - u_h\|_{0,p,\Omega} \le C\|\sigma - \Pi_h\sigma\|_{0,q,\Omega}^{q/p}.$$

Finally, error estimate (3.9) is a consequence of this last one, (3.6), the triangle inequality and the following standard error estimation

$$\|u - P_h u\|_{0,p,\Omega} \le Ch|u|_{1,p,\Omega}. \qquad \square$$

Remark 3.2. In the proof of Theorem 3.1, we have used the fact that $\|\sigma_h\|_{0,q,\Omega} \le C$ with C a constant independent of h. The proof of this bound is analogous to the one for σ in Theorem 2.1, due to the inf-sup condition (3.7).

The error estimation (3.9) is the same as the one obtained in Glowinski and Marrocco [**8**], while the error estimation (3.8) is better than the one obtained in Glowinski and Marrocco [**8**].

In the case $1 < p < 2$, we get better error estimates. Furthermore, these error estimates are valid without restrictions on p, i.e., for all $p \in]1, 2[$. This is the main difference between Theorem 3.2 in [**6**] and the following one.

Theorem 3.2. *Suppose that* $1 < p < 2$. *Let* (σ, u) *be the solution of* (2.2) *and* (σ_h, u_h) *the solution of* (3.3). *If* $(\sigma, u) \in (W^{1,q}(\Omega))^N \times W^{1,p}(\Omega)$, *then there exists a constant* C *independent of* h *such that*

$$\|\sigma - \sigma_h\|_{0,q,\Omega} \leq Ch^{2/q}, \tag{3.12}$$

$$\|u - u_h\|_{0,p,\Omega} \leq Ch. \tag{3.13}$$

Proof. As in the proof of Theorem 3.1, we have

$$\int_\Omega (A(\sigma) - A(\sigma_h)) \cdot (\sigma - \sigma_h)\, dx = \int_\Omega (A(\sigma) - A(\sigma_h)) \cdot (\sigma - \Pi_h\sigma)\, dx.$$

Now, from (cf. Sandri [**17**])

$$\begin{aligned} \int_\Omega (A(\sigma) - A(\sigma_h)) \cdot (\sigma - \sigma_h)\, dx \\ \geq C\{\|\sigma - \sigma_h\|^q_{0,q,\Omega} + \int_\Omega (|\sigma| + |\sigma_h|)^{q-2} |\sigma - \sigma_h|^2\, dx\}, \end{aligned} \tag{3.14}$$

$$\begin{aligned} \|A(\sigma) - A(\sigma_h)\|_{0,p,\Omega} \leq C\bigg[\int_\Omega (|\sigma| + |\sigma_h|)^{q-2} |\sigma - \sigma_h|^2\, dx\bigg]^{1/2} \\ \cdot [\|\sigma\|_{0,q,\Omega} + \|\sigma_h\|_{0,q,\Omega}]^{(q-2)/2}, \end{aligned} \tag{3.15}$$

and using Young's inequality, we obtain for all $\varepsilon > 0$,

$$\begin{aligned} \|\sigma - \sigma_h\|^q_{0,q,\Omega} + \int_\Omega (|\sigma| + |\sigma_h|)^{q-2} |\sigma - \sigma_h|^2\, dx \\ \leq \varepsilon \int_\Omega (|\sigma| + |\sigma_h|)^{q-2} |\sigma - \sigma_h|^2\, dx \\ + \frac{C}{\varepsilon} [\|\sigma\|_{0,q,\Omega} + \|\sigma_h\|_{0,q,\Omega}]^{q-2} \|\sigma - \Pi_h\sigma\|^2_{0,q,\Omega}, \end{aligned}$$

where C is a constant independent of h and ε. Using this last estimate and Remark 3.2, we get

$$\|\sigma - \sigma_h\|_{0,q,\Omega}^q + \int_\Omega (|\sigma| + |\sigma_h|)^{q-2} |\sigma - \sigma_h|^2 \, dx \leq C \|\sigma - \Pi_h \sigma\|_{0,q,\Omega}^2. \tag{3.16}$$

Then (3.12) is a consequence of this last estimation and (3.6).

For the error estimation (3.13) we have (see the proof of Theorem 3.1)

$$\|P_h u - u_h\|_{0,p,\Omega} \leq C \|A(\sigma) - A(\sigma_h)\|_{0,p,\Omega},$$

and, using (3.15) and (3.16), we obtain

$$\|P_h u - u_h\|_{0,p,\Omega} \leq C \|\sigma - \Pi_h \sigma\|_{0,q,\Omega}.$$

Therefore, (3.13) is a consequence of this last inequality, (3.6), the triangle inequality and the standard error estimate on the operator P_h. □

Remark 3.3. The error estimate (3.12) for σ is better than the one obtained in Baranger et al. [**2**] for a quasi-Newtonian flow, and the error estimation (3.13) for u is optimal. Furthermore, these error estimates are valid with minimal regularity of the solution of problem (1.1) (compare with Theorem 3.2 in [**6**], and the results obtained in Barrett and Liu [**3**]).

REFERENCES

1. C. Atkinson and C.R. Champion, *Some boundary-value problems for the equation* $\nabla \cdot (|\nabla\phi|^N \nabla\phi) = 0$, Quart. J. Mech. Appl. Math. **37** (1984), 401–419.

2. J. Baranger, K. Najib and D. Sandri, *Numerical analysis of a three-fields model for a quasi-Newtonian flow*, Comput. Methods Appl. Mech. Engrg. **109** (1993), 281–292.

3. J.W. Barrett and W.B. Liu, *Finite element approximation of the p-Laplacian*, Math. Comp. **61** (1993), 523–537.

4. F. Brezzi and M. Fortin, *Mixed and hybrid finite element methods*, Springer-Verlag, New York, 1991.

5. P.G. Ciarlet, *The finite element method for elliptic problems*, North-Holland, Amsterdam, 1978.

6. M. Farhloul, *A mixed finite element method for a nonlinear Dirichlet problem*, IMA J. Numer. Anal. **18** (1998), 121–132.

7. V. Girault and P.A. Raviart, *Finite element for Navier-Stokes equations, theory and algorithms*, Springer Ser. Comput. Math. **5**, Springer-Verlag, New York, 1986.

8. R. Glowinski and A. Marrocco, *Sur l'approximation par éléments finis d'ordre un, et la résolution, par pénalisation-dualité, d'une classe de problèmes de Dirichlet non linéaires*, RAIRO, R-2, 9e année (1975), 41–76.

9. P. Le Tallec, *Numerical analysis of viscoelastic problems*, Rech. Math. Appl. **15**, Masson, Springer-Verlag, 1990.

10. W.B. Liu and J.W. Barrett, *Quasi-norm error bounds for the finite element approximation of some degenerate quasilinear elliptic equations and variational inequalities*, M^{2}AN **28** (1994), 725–744.

11. ———, *Finite element approximation of some degenerate monotone quasilinear elliptic systems*, SIAM J. Numer. Anal. **33** (1996), 88–106.

12. H. Manouzi and M. Farhloul, *Mixed finite element analysis of a non-linear three-fields Stokes model*, IMA J. Numer. Anal., to appear.

13. J.C. Nedelec, *Mixed finite elements in* $\mathbf{R}^3$, Numer. Math. **35** (1980), 315–341.

14. J.R. Philip, *N-diffusion*, Austral. J. Phys. **14** (1961), 1–13.

15. P.A. Raviart and J.M. Thomas, *A mixed finite element method for* 2*-nd order elliptic problems*, Lecture Notes in Math. **606** (1977), 292–315.

16. J.E. Roberts and J.M. Thomas, *Mixed and hybrid finite element methods*, in *Handbook of numerical analysis*, Vol. II, *Finite element methods*, Part 1 (P.G. Ciarlet and J.L. Lions, eds.), North-Holland, Amsterdam, 1989.

17. D. Sandri, *Sur l'approximation numérique des écoulements quasi-Newtoniens dont la viscosité suit la loi puissance oú la loi de Carreau*, M^{2}AN **27** (1993), 131–155.

18. J.M. Thomas, *Sur l'analyse numérique des méthodes d'éléments finis hybrides et mixtes*, Thèse, Université Pierre et Marie Curie, Paris, 1977.

19. V.B. Tyukhtin, *Sur la vitesse de convergence des méthodes d'approximation de la solution des problèmes variationnels unilatéraux*, Vestnik Leningrad Univ., Math. Mec. Astronom. **3** (1983), 36–43 (en russe).

Département de Mathématiques et de Statistique, Université de Moncton, Moncton, NB, E1A 3E9 Canada
E-mail address: farhlom@umoncton.ca

Département de Mathématiques et de Statistique, Université Laval, Québec, G1K 7P4 Canada

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